

Task 3 — Car Price Prediction

Description

Three regression models — **Linear Regression**, **Random Forest**, and **Gradient Boosting** — are trained to predict the selling price of used cars.

Dataset

Loaded automatically from GitHub:

https://raw.githubusercontent.com/RimjimRazdan/cars_price_prediction/master/car%20data.csv

Features

- Price distribution analysis (normal + log-transformed)
- Correlation heatmap
- Categorical analysis: Fuel Type & Seller Type
- Feature engineering: $\text{Car_Age} = 2024 - \text{Year}$
- Label encoding for categorical columns
- 5-fold cross-validation

Models Used

Model	Parameters
Linear Regression	Default
Random Forest	<code>n_estimators=200</code>
Gradient Boosting	<code>n_estimators=200</code>

Evaluation Metrics

- MAE (Mean Absolute Error)
- RMSE (Root Mean Squared Error)
- R^2 Score
- Cross-Validation R^2

Output Files

- `car_price_distribution.png`
- `car_correlation.png`
- `car_categorical_analysis.png`
- `car_model_comparison.png`
- `car_actual_vs_predicted.png`
- `car_feature_importance.png`

Quick Start

bash

```
pip install numpy pandas matplotlib seaborn scikit-learn  
python task3_car_price.py
```

output

=== Dataset Info ===

	Car_Name	Year	Selling_Price	Present_Price	Kms_Driven	Fuel_Type	\
0	ritz	2014	3.35	5.59	27000	Petrol	
1	sx4	2013	4.75	9.54	43000	Diesel	
2	ciaz	2017	7.25	9.85	6900	Petrol	
3	wagon r	2011	2.85	4.15	5200	Petrol	
4	swift	2014	4.60	6.87	42450	Diesel	

	Seller_Type	Transmission	Owner
0	Dealer	Manual	0
1	Dealer	Manual	0
2	Dealer	Manual	0
3	Dealer	Manual	0
4	Dealer	Manual	0

Shape: (301, 9)

Column Types:

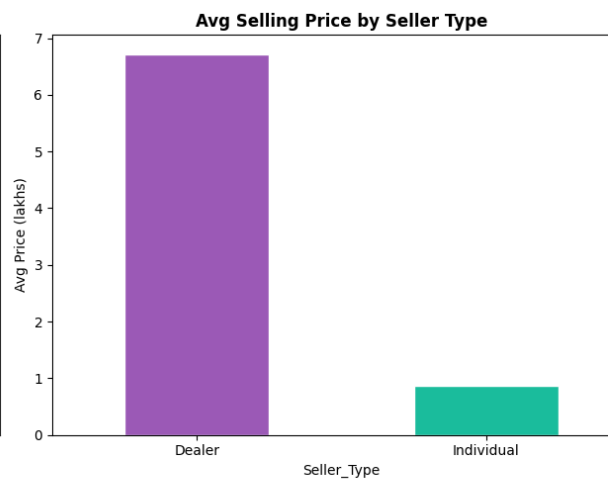
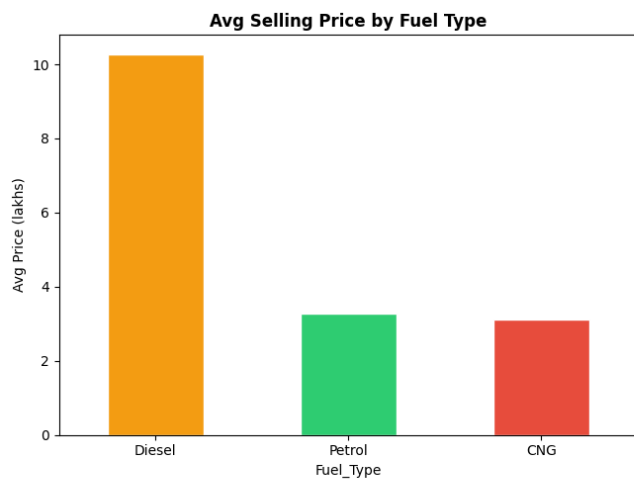
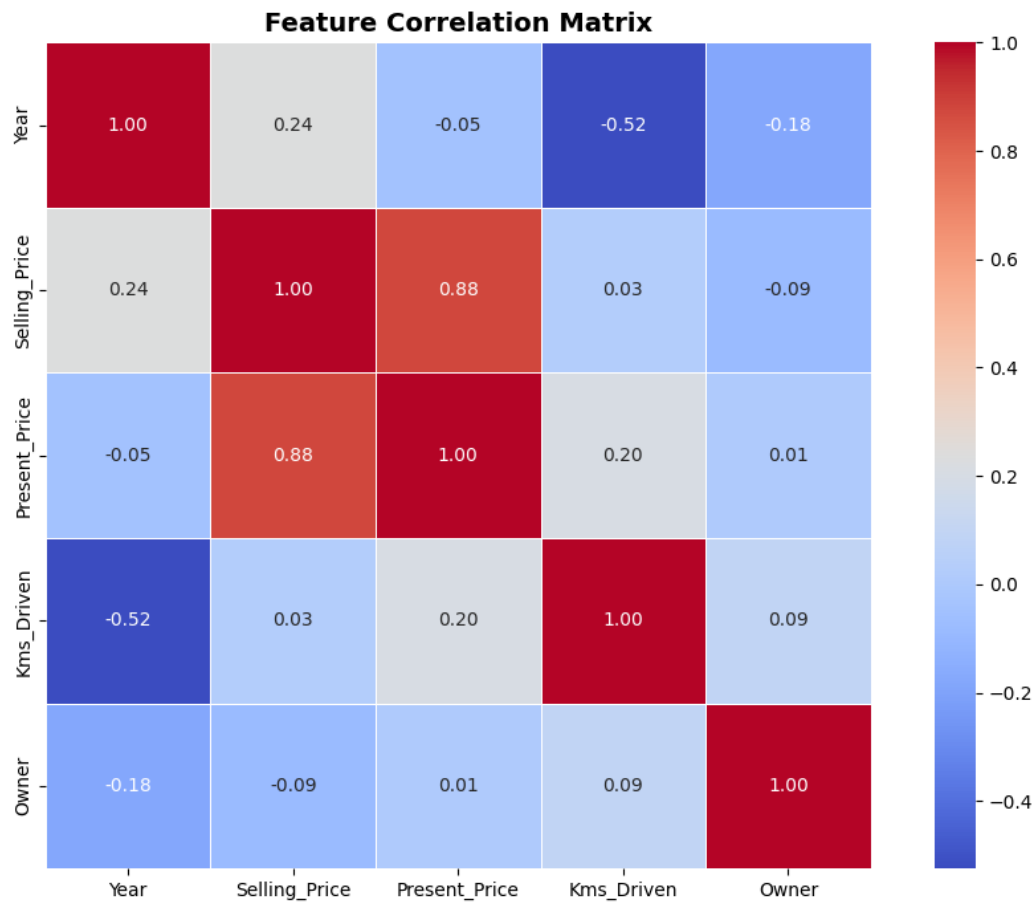
Car_Name	object
Year	int64
Selling_Price	float64
Present_Price	float64
Kms_Driven	int64
Fuel_Type	object
Seller_Type	object
Transmission	object
Owner	int64

dtype: object

Null Values:

Car_Name	0
Year	0
Selling_Price	0
Present_Price	0
Kms_Driven	0
Fuel_Type	0
Seller_Type	0
Transmission	0
Owner	0

dtype: int64



```

=== Processed Dataset ===
  Selling_Price  Present_Price  Kms_Driven  Fuel_Type  Seller_Type  \
0          3.35           5.59        27000         2           0
1          4.75           9.54        43000         1           0
2          7.25           9.85         6900         2           0
3          2.85           4.15         5200         2           0
4          4.60           6.87        42450         1           0
  
```

	Transmission	Owner	Car_Age
0	1	0	10
1	1	0	11
2	1	0	7
3	1	0	13
4	1	0	10

=== Linear Regression ===

MAE : 1.2218

RMSE : 1.8785

R² : 0.8468

CV R²: 0.8446

=== Random Forest ===

MAE : 0.6214

RMSE : 0.9549

R² : 0.9604

CV R²: 0.8800

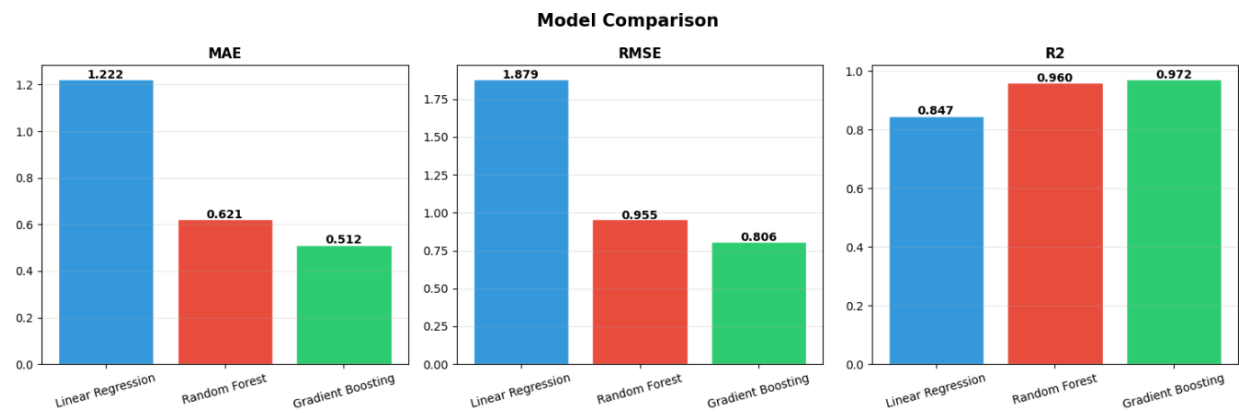
=== Gradient Boosting ===

MAE : 0.5118

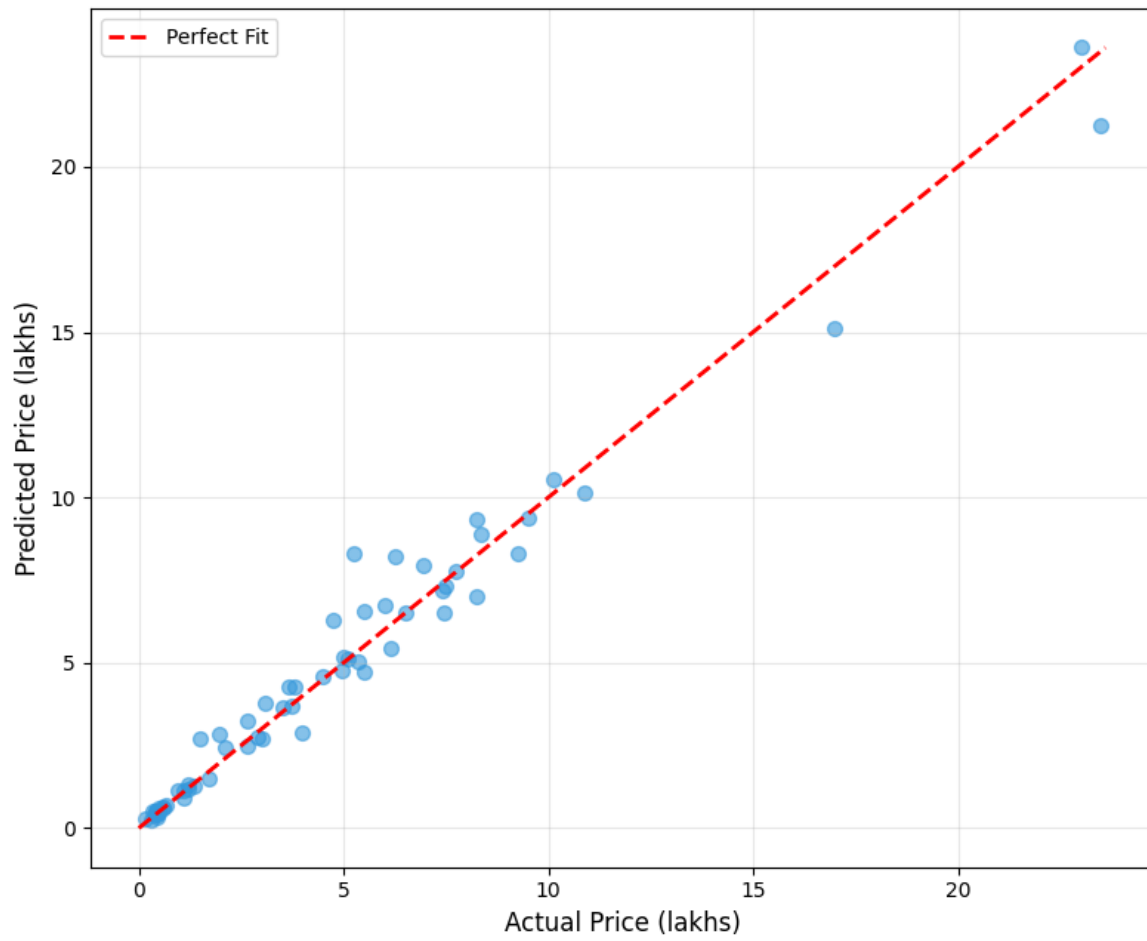
RMSE : 0.8061

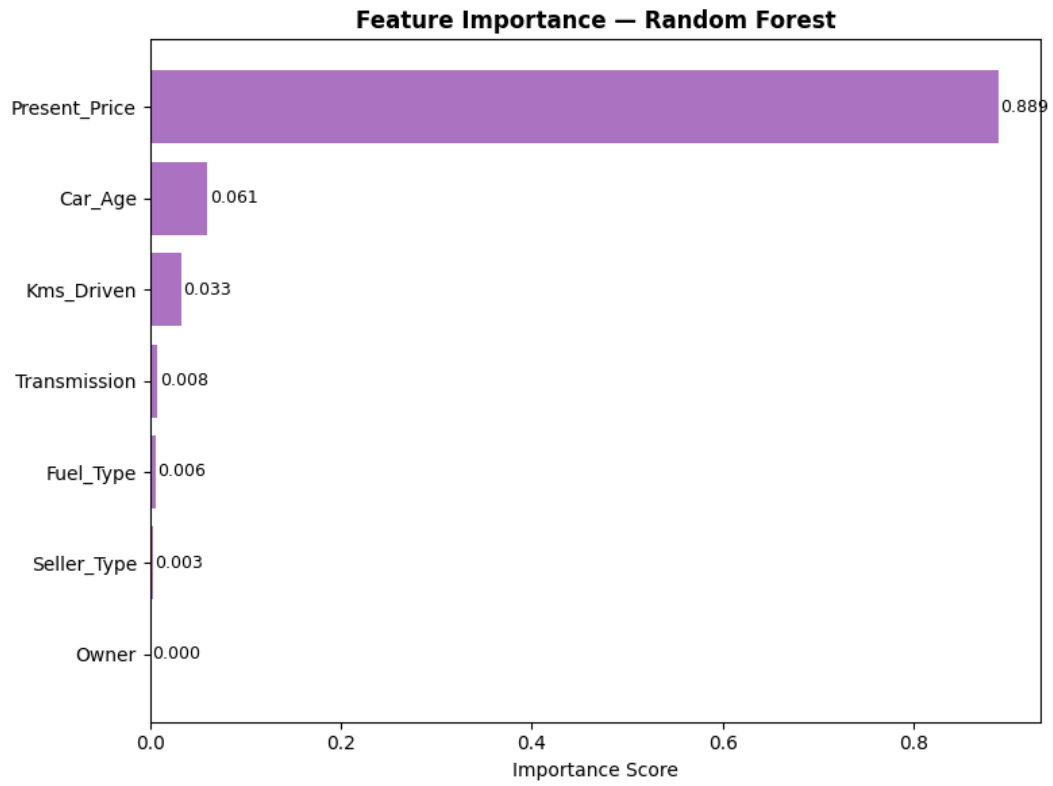
R² : 0.9718

CV R²: 0.8910



Actual vs Predicted — Gradient Boosting
 $R^2 = 0.9718$





✔ Predicted Selling Price for sample car: ₹ 5.56 lakhs